MAT2540, Classwork15, Spring2026

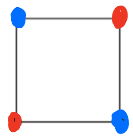
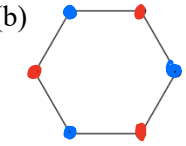
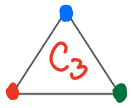
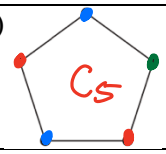
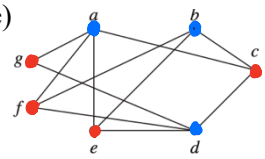
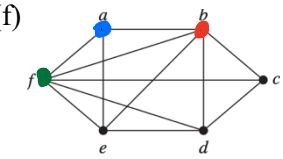
10.2 Graph Terminology and Special Types of Graphs (Conti.)

13. Definition: bipartite

A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a bipartition of the vertex set V of G.

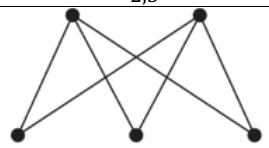
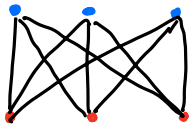
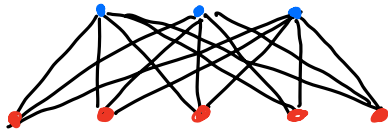
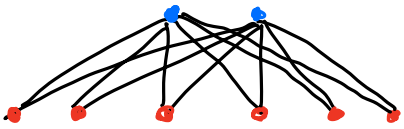
14. *Theorem*. If one uses the same color to draw any non-adjacent vertices from a graph, Then a graph is bipartite if and only if it is 2-colorable.

15. *Examples*. Check if each given graph is a bipartite or not.

(a)  Yes, it is a bipartite	(b)  Yes, it is a bipartite
(c)  C_3 NO, it is NOT a bipartite	(d)  C_5 NO, it is not a bipartite
(e)  Yes, it is a bipartite	(f)  NO, it is not a bipartite

16. *Theorem*. An undirected graph is bipartite if and only if it does not contain an odd cycle. C_n, n is odd number

17. *Complete Bipartite Graphs*. A complete bipartite graph $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.

$K_{2,3}$	$K_{3,3}$	$K_{3,5}$	$K_{2,6}$
			

18. Theorem. An undirected graph has an even number of vertices of odd degree. (why?) 3, 3, 3, 2, 2, 2 sum of degree has to be even

19. A sequence $d_1, d_2, \ldots, d_n$ is called **graphic** if it is the degree sequence of a simple graph. Determine whether each of these sequences is graphic.

(a) 5, 4, 3, 2, 1, 0 (b) 2, 2, 2, 2, 2, 2 (c) 1, 1, 1, 1, 1, 1

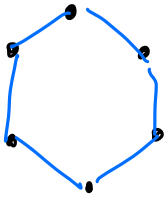
$5+4+3+2+1+0 = 15$

$\frac{15}{2}$ is NOT an integer

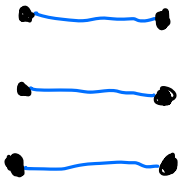
NOT a graph

$\frac{2+2+2+2+2+2}{2} = 6$

Yes, it is a graph



$\frac{1+1+1+1+1+1}{2} = 3$



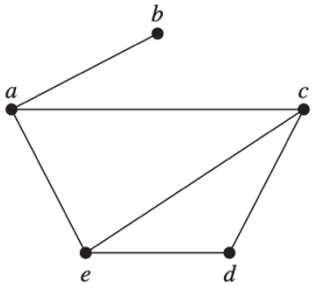
10.3 Representing Graphs and Graph Isomorphism

1. *Definition:* Adjacency Matrix based on the Adjacency of Vertices of an **Undirected Graph**.

Suppose that $G = (V, E)$ is an **undirected graph** where $|V| = n$. Suppose that $v_1, v_2, \ldots, v_n$ are the vertices of G. The adjacency matrix A (or A_G) of G, with respect to this listing of the vertices, is the $n \times n$ zero–one matrix with m as its (i, j)th entry when v_i and v_j are adjacent with multiplicity m, and 0 as its (i, j)th entry when they are adjacent. In other words, if its adjacency matrix is $A = [a_{ij}]$, then

$$a_{ij} = \begin{cases} m & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise}. \end{cases}$$

2. *Representation of Graph:* Adjacency List and Adjacency Matrix For a Simple Graph.



Vertex	Adjacent Vertices	deg.
a	b, c, e	3
b	a	1
c	a, d, e	3
d	c, e	2
e	a, c, d	3

column

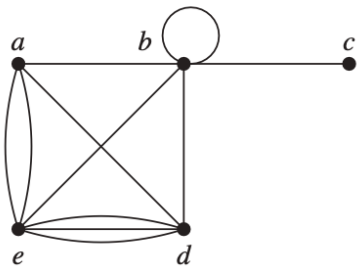
row

	a	b	c	d	e	
a	0	1	1	0	1	3
b	1	0	0	0	0	1
c	1	0	0	1	1	3
d	0	0	1	0	1	2
e	1	0	1	1	0	3
	3	1	3	2	3	

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

diagonal

3. *Representation of Graph:* Adjacency List and Adjacency Matrix For a Multigraph.



Vertex	Adjacent Vertices	deg.
a		
b		
c		
d		
e		

	a	b	c	d	e
a					
b					
c					
d					
e					

$$\begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix}$$

4.Observations of the Adjacency Matrix For an Undirected Graph

Loops:

Sum of the row/column:

Symmetric property:

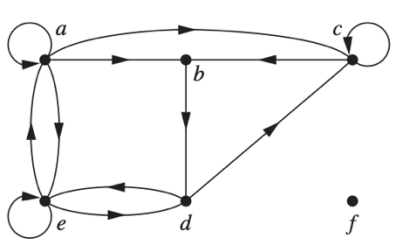
5. *Definition:* Adjacency Matrix based on the Adjacency of Vertices of a **Directed Graph**.

Suppose that $G = (V, E)$ is a **directed graph** where $|V| = n$. Suppose that $v_1, v_2, \ldots, v_n$ are the vertices of G. The adjacency matrix A (or A_G) of G, with respect to this listing of the vertices, is the $n \times n$ zero–one matrix with ____ as its (i, j)th entry when _____, and _____.

In other words, if its adjacency matrix is $A = [a_{ij}]$, then

$$a_{ij} = \begin{cases} m & \text{if } (v_i, v_j) \text{ is a directed edge of } G, \\ 0 & \text{otherwise}. \end{cases}$$

6. *Representation of Graph:* Adjacency Matrix For a Directed Graph.



	In-deg.	Out-deg.
a		
b		
c		
d		
e		
f		

	a	b	c	d	e	f
a						
b						
c						
d						
e						
f						

$$\begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}$$